Lecture 18
Sampling Distributions, the Central Limit Theorem and Confidence Intervals

Review

1 What is a Poisson random variable?

The number of events in a fixed interval, when events occur independently at an average rate λ.

2 What is a probability density?

How concentrated probability is around a point; probability is the area under the curve.

3 For a continuous random variable, what is P(X = c)?

0: a single value has no width, so it has no area under the curve.

4 What is a cumulative probability?

$P(X \leq x)$: the area to the left of x, what the z-table and pnorm give.

Two polls, two answers

Two polling firms each survey a random sample of 1,000 voters in the same state, in the same week.

Poll A: $\hat{p} = 0.52$ Poll B: $\hat{p} = 0.49$

(a) Is one of the polls wrong?

(b) If 1,000 firms each polled 1,000 voters, what would their 1,000 values of $\hat{p}$ look like?

(a) Not necessarily: different random samples give different values of $\hat{p}$

(b) A pile of values centered near p: the sampling distribution of $\hat{p}$

Three distributions

population distribution

how a single observation X varies; described by parameters such as μ, σ and p

data distribution

the n values in our one sample; described by statistics such as $\bar{x}$, s and $\hat{p}$

sampling distribution

how a statistic such as $\bar{x}$ varies over all possible samples of the same size n

Building a sampling distribution

A spinner lands on 1, 2 or 3, each with probability 1/3. Spin it twice and record the mean, $\bar{x}$.

$\mu = 2$; n = 2 spins; $3 \times 3 = 9$ equally likely samples (order matters)

(a) List the 9 samples and the mean of each.

(b) What is $P(\bar{x})$ for each possible value of $\bar{x}$?

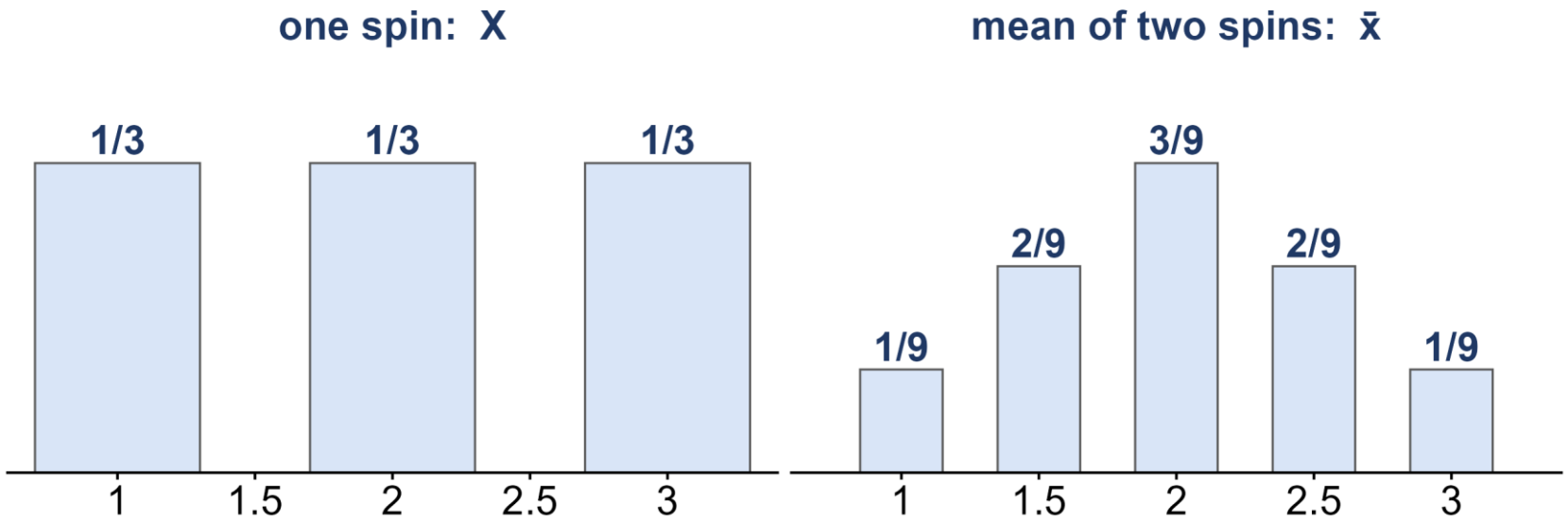
(a) (1,1) → 1; (1,2), (2,1) → 1.5; (1,3), (2,2), (3,1) → 2; (2,3), (3,2) → 2.5; (3,3) → 3

(b) $\bar{x}$ = 1, 1.5, 2, 2.5, 3 with $P(\bar{x})$ = 1/9, 2/9, 3/9, 2/9, 1/9

One spin vs. the mean of two spins

The population (left) and the sampling distribution of $\bar{x}$ for n = 2 (right).

What stayed the same? What changed?



Same center: both have mean 2. Smaller spread: $\sigma = 0.82$, $\sigma_{\bar{x}} = 0.58 = 0.82 / \sqrt{2}$
New shape: peaked in the middle

Center and spread of $\bar{x}$

mean of $\bar{x}$

$\mu_{\bar{x}} = \mu$: $\bar{x}$ is centered at the parameter it estimates

standard error of $\bar{x}$

$\sigma_{\bar{x}} = \sigma / \sqrt{n}$: the spread of $\bar{x}$, which shrinks as n grows

the sample proportion $\hat{p}$

mean p and standard error $\sigma_{\hat{p}} = \sqrt{(p(1 - p) / n)}$

Data: Spotify songs

Spotify and YouTube statistics for the top 10 songs of each of 2,079 popular artists, collected in February 2023.

X = the length of a song, in minutes

TRACK	ARTIST	MINUTES
Feel Good Inc.	Gorillaz	3.71
Rhinestone Eyes	Gorillaz	3.34
New Gold (feat. Tame Impala and Bootie Brown)	Gorillaz	3.59
On Melancholy Hill	Gorillaz	3.90
Clint Eastwood	Gorillaz	5.68
…	…	…

18,860 different songs (a song credited to several artists is counted once)

Example: standard error and n

Treat the 18,860 songs as the population. Draw a random sample of n songs and average their lengths to get $\bar{x}$. Every new sample gives a different $\bar{x}$; the standard error $\sigma_{\bar{x}}$ measures how much $\bar{x}$ varies from sample to sample.

μ = 3.71 min; σ = 1.51 min

(a) What is $\sigma_{\bar{x}}$ for n = 20?
(b) What is $\sigma_{\bar{x}}$ for n = 80?
(c) How much larger must a sample be to cut the standard error in half?

(a) 1.51 / √20 ≈ 0.34 min (b) 1.51 / √80 ≈ 0.17 min
(c) 4 times larger, because √4 = 2

Data: waiting for a goal

The 2022 World Cup had 172 goals in 64 matches: 2.69 goals per 90-minute match.

X = minutes until the next goal, simulated at 2.69 goals per 90 minutes

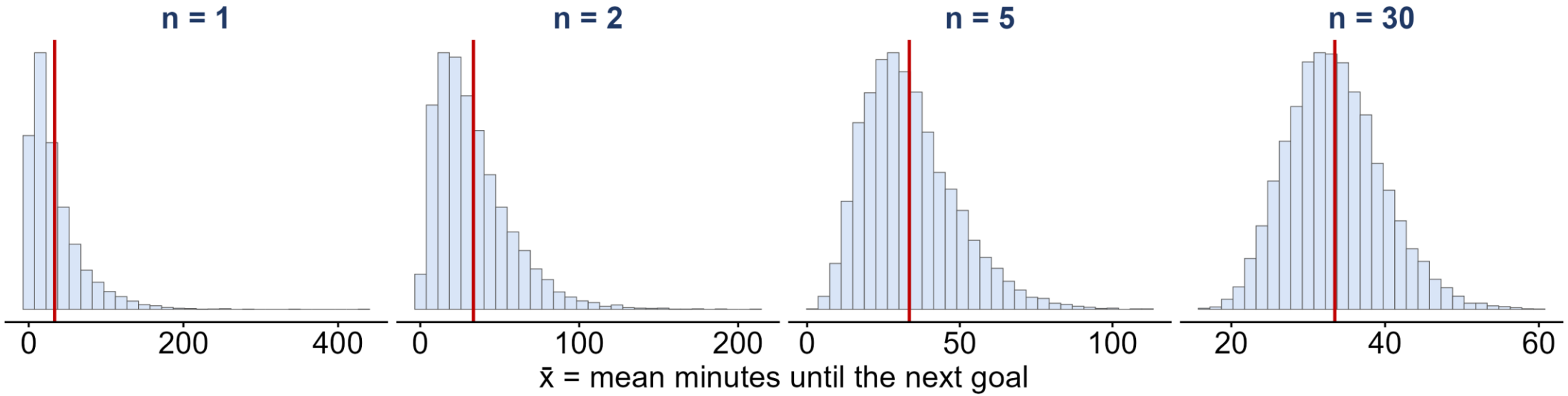
WAIT	MINUTES
1	1.6
2	54.3
3	57.3
4	27.2
5	18.3
…	…

$\mu = 90 / 2.69 \approx 33.5$ min; $\sigma \approx 33.5$ min. Most waits are short and a few are very long.

Averaging a skewed variable

Each panel: draw n waits, average them to get $\bar{x}$, and repeat 10,000 times (n = 1 is a single wait).

As n grows, what happens to the shape, the center and the spread of $\bar{x}$?



Shape: the skew fades into a bell. Center: stays at 33.5. Spread: shrinks like σ / √n

The central limit theorem

central limit theorem

for large n, $\bar{x}$ is approximately $N(\mu, \sigma / \sqrt{n})$, whatever the shape of the population

how large is large enough?

$n \geq 30$ for $\bar{x}$; $np > 5$ and $n(1 - p) > 5$ for $\hat{p}$

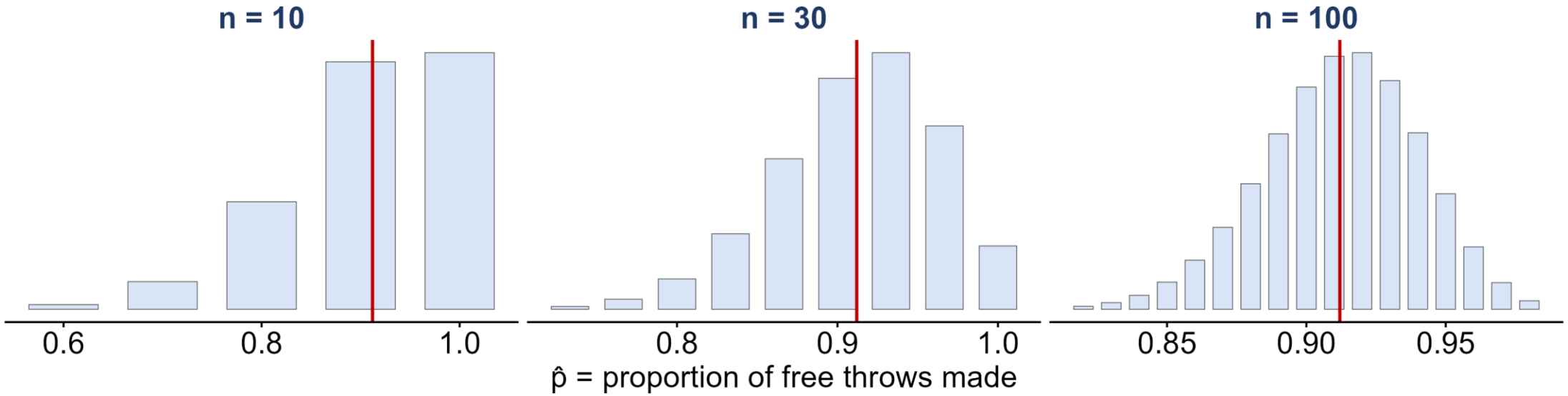
why it matters

the empirical rule applies: about 95% of samples give a statistic within 2 SE of its mean

The CLT for a proportion

Exact sampling distributions of $\hat{p}$, the proportion of free throws Stephen Curry makes; $p = 0.912$.

For which n is $n(1 - p) > 5$? Which distributions look like a bell?



$n(1 - p) = 0.88,\ 2.64,\ 8.8$: only n = 100 passes, and only n = 100 is close to a bell

Example: free throws, again

Over a stretch of n = 100 free throws, let $\hat{p}$ be the proportion Curry makes.

$p = 0.912$; $n = 100$; $n(1 - p) = 8.8 > 5$

(a) What are the mean and standard error of $\hat{p}$?

(b) About 95% of the time, $\hat{p}$ falls between what two values?

(a) mean 0.912; SE = $\sqrt{(0.912 \times 0.088 / 100)} \approx 0.028$

(b) $0.912 \pm 2(0.028)$: 0.856 to 0.968

Example: a tournament's average

Goals per World Cup match: $X \sim \text{Poisson}(\lambda = 2.69)$. A tournament averages over 64 matches.

$\mu = \lambda = 2.69$; $\sigma = \sqrt{2.69} \approx 1.64$; $n = 64$

(a) What is the standard error of a tournament's average?
(b) About 95% of tournaments average between what two values?
(c) The 2018 World Cup averaged 169 / 64 = 2.64 goals per match. Surprising?

(a) $1.64 / \sqrt{64} \approx 0.205$
(b) $2.69 \pm 2(0.205)$: 2.28 to 3.10 goals per match
(c) No: 2.64 is well inside 2.28 to 3.10

From an estimate to an interval

point estimate

a single statistic used to estimate a parameter: $\bar{x}$ for μ, $\hat{p}$ for p

confidence interval

a range of plausible values for the parameter: estimate ± margin of error

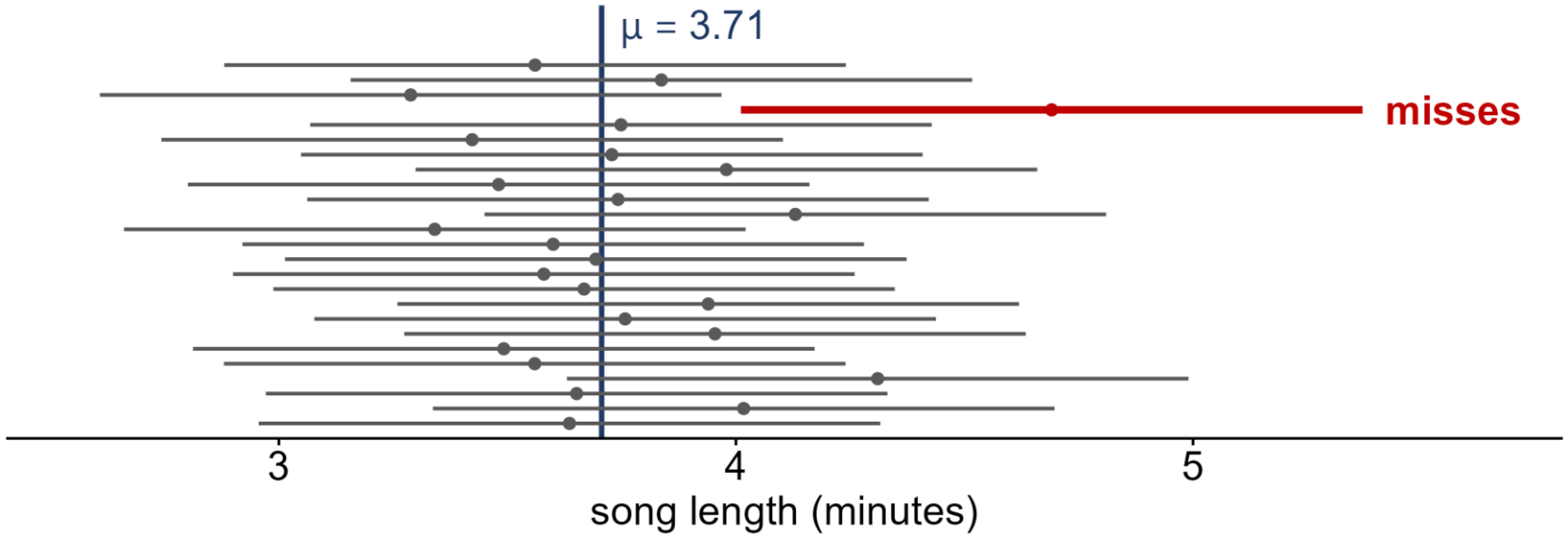
95% confidence

the method captures the parameter in about 95% of all possible samples

How often does an interval capture μ?

25 random samples of n = 20 songs; each line is $\bar{x} \pm 2(0.34)$, the margin of error.

How many of the 25 intervals capture μ = 3.71?



24 of 25: about 95% of intervals made this way capture μ

Example: an interval for mean song length

In practice we see only one sample: n = 20 songs (the bottom interval on the last slide).

$\bar{x}$ = 3.64 min; σ = 1.51 min; SE = 0.34 min

(a) What is the 95% confidence interval for μ?
(b) Does it capture μ = 3.71?
(c) If we did not know μ, what would "95% confident" mean?

(a) 3.64 ± 2(0.34): 2.96 to 4.32 min (b) Yes
(c) 95% of intervals built this way capture μ; this one either does or does not

What you should know

sampling distribution

the distribution of a statistic over all possible samples of size n

standard error

$\sigma_{\bar{x}} = \sigma / \sqrt{n}$ and $\sigma_{\hat{p}} = \sqrt{(p(1 - p) / n)}$: both shrink as n grows

central limit theorem

for large n, $\bar{x}$ and $\hat{p}$ are approximately normal, whatever the population

confidence interval

estimate ± 2 SE: captures the parameter for about 95% of samples